LARGE SAMPLE TESTS

TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 14

5-MINUTE REVISION

Source: Chapter 14 of the textbook.

Book citation: Pratheesh P. Gopinath, Adarsh V.S., Mohammed Hisham M., & Manju Mary Paul. (2026). *Textbook of Agricultural Statistics*. Statoberry LLP. Chapter 14.

1. QUICK REVISION · 1

- Large sample → generally $n \geq 30$.
- Large-sample tests are based on the Central Limit Theorem.
- Central Limit Theorem → sampling distribution of $\bar{X}$ approaches a normal distribution as n increases.
- The observations should be independent.
- The population should have finite variance.

2. QUICK REVISION · 2

- Large-sample Z-test → based on $N(0, 1)$.
- Tests in this chapter → single proportion, equality of two proportions, single mean, equality of two means, and an approximate comparison of two standard deviations under the stated normal-population assumptions.
- Single proportion → compare P with a specified value P_0.
- For a single proportion, $H_0 : P = P_0$.
- For a single proportion, the alternative may be $P < P_0$, $P > P_0$, or $P \neq P_0$.

3. QUICK REVISION · 3

- Single-proportion standard error uses P_0 and Q_0 under the null hypothesis.
- Two proportions → compare P_1 and P_2.
- For two proportions, $H_0 : P_1 = P_2$.
- The two-proportion test uses the pooled proportion $\widehat{P}$.
- Single mean → compare μ with μ_0.

4. QUICK REVISION · 4

- When σ is known, use $\sigma/\sqrt{n}$.
- When σ is unknown, use $s/\sqrt{n}$.
- Equality of two means → compare μ_1 and μ_2.
- Equal population standard deviations → use the common σ.
- Equal but unknown standard deviations → use the pooled estimate.

5. QUICK REVISION · 5

- Unequal population standard deviations → use separate variance terms.
- Two-tailed test → reject if $|Z| > Z_{\alpha/2}$.
- Right-tailed test → reject if $Z > Z_{\alpha}$.
- Left-tailed test → reject if $Z < -Z_{\alpha}$.
- At $\alpha = 0.05$, two-tailed critical value → 1.960.

6. QUICK REVISION · 6

- At $\alpha = 0.01$, two-tailed critical value → 2.576.
- At $\alpha = 0.05$, right-tailed critical value → 1.645.
- At $\alpha = 0.05$, left-tailed critical value → -1.645.
- The null hypothesis is always stated in terms of the population parameter.
- The test statistic is calculated from sample statistics.

7. QUICK REVISION · 7

- If the calculated statistic falls in the critical region, reject H_0.
- Otherwise, do not reject H_0.
- For a two-tailed test, α is divided equally between the two tails.
- The sign of Z is important in one-tailed tests; $|Z|$ is used for the two-tailed decision.
- A large absolute value of Z indicates greater departure from the null hypothesis.

8. IMPORTANT FORMULAE · 1

Central Limit Theorem:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

Single population proportion:

$$Z = \frac{p - P_0}{\sqrt{P_0 Q_0 / n}}$$

Complementary proportion:

$$Q_0 = 1 - P_0$$

9. IMPORTANT FORMULAE · 2

Pooled proportion for two proportions:

$$\widehat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

Pooled complementary proportion:

$$\widehat{Q} = 1 - \widehat{P}$$

Equality of two proportions:

$$Z = \frac{p_1 - p_2}{\sqrt{\widehat{P}\widehat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

10. IMPORTANT FORMULAE · 3

Single population mean when σ is known:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Single population mean when σ is unknown:

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Equality of two means when population standard deviations are equal:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

11. IMPORTANT FORMULAE · 4

Pooled estimate of common standard deviation:

$$\sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}}$$

Equality of two means when population standard deviations are unequal:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Difference of two standard deviations (population standard deviations unknown):

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}}$$

12. SELF-CHECK FROM THE CHAPTER

1. The theorem supporting the normal approximation to the sample mean under suitable conditions is the _____ Limit Theorem. **[Central]**
2. The standard deviation of a sampling distribution is called the standard _____. **[Error]**
3. The reference distribution used in a Z-test is the standard _____ distribution. **[Normal]**
4. For independent observations with population standard deviation σ, the standard error of the sample mean is _____. **[$\sigma/\sqrt{n}$]**